

Formulas

set of classes:	C
set of samples:	S
feature vectors:	$x_s \in \mathbb{R}^n$
weights:	$\theta_c \in \mathbb{R}^n$
labels:	$y_{s,c} \in \{0, 1\}$
loss function:	$L(x, y) = -xy + \ln(1 + e^x)$
regularizer:	$R(\theta_c) = \frac{\lambda}{2} \ \theta_c\ ^2$
goal:	$\min \sum_{c \in C} R(\theta_c) + \sum_{s \in S} L(\langle \theta_c, x_s \rangle, y_{s,c})$
constraints:	$\forall s \in S : \sum_{c \in C} y_{s,c} = 1$ $\forall c \in C : \sum_{s \in S} y_{s,c} = \frac{ S }{ C }$

ILP

$$\begin{aligned} & \text{given } \theta_c \text{ and } x_s \\ & \min \sum_{c \in C} R(\theta_c) + \sum_{s \in S} L(\langle \theta_c, x_s \rangle, y_{s,c}) \\ & \min \sum_{c \in C} \sum_{s \in S} -\langle \theta_c, x_s \rangle y_{s,c} \end{aligned}$$

GD

given $y_{s,c}$ and x_s

$$\forall c \in C : \nabla_{\theta_c} \left(R(\theta_c) + \sum_{s \in S} L(\langle \theta_c, x_s \rangle, y_{s,c}) \right) = \lambda \theta_c + \sum_{s \in S} x_s \left(-y_{s,c} + \frac{1}{e^{-\langle \theta_c, x_s \rangle} + 1} \right)$$